Ethosuximide

Card EB-ETX-01 · Aim: useful partial benefit · Track A (variant-independent) · status: hypothesis · Group 1

Data date: 2026-10-28 (needs record 2026-10-27; catalogue build 2026-10-28T07:32:18Z).
Perimeter: WWOX-related developmental and epileptic encephalopathy (WWOX-DEE), reference genotype class; public literature and public databases only; no individual-level data. Research material for discussion with a treating clinical team: not medical advice, not a treatment recommendation, not a probability of benefit.

Rationale

Origin: the Wwox-null mouse PTZ experiment (WWOX-direct), not another disease. Transfer to human WOREE: untested; one experiment, one laboratory, symptomatic class at best.

Needs addressed: Seizure and spasm control as a symptom-level need; Ethosuximide sensitivity of provoked seizure in Wwox loss.

Evidence, by scope

- ● WWOX experiment in a DEE system PTZ-provoked behavioural seizure score, late phase of the hour (graphical reading): decrease – R2 reading: the effect is confined to the second half of the hour, the PTZ-alone null control varies between the ethosuximide and the lithium experiments by a comparable order to the late effect (the Methods do not say whether the cohorts are separate), and … (PMID 32000863)
- – Scope not recorded at assertion level PTZ-provoked behavioural seizure score, late phase of the hour (graphical reading): unknown – Neither a protective effect nor a clear absence. Whether the contrast is a floor effect, a baseline effect or a real null-specific effect is not distinguished by these data.
- ▲ Other-disease precedent viability of embryonic cortical progenitors in vitro: decrease – Hypothesis-generating class-level safety signal only: in vitro, research compounds, Cav3.1-dependent; in vivo and clinical significance unknown; not an administration statement and not a statement about ethosuximide.
- *Observations recorded on the linked need(s): context only, they may concern other interventions.*
- ● WWOX experiment in a DEE system *need-level observation:* In the Wwox-null mouse, ethosuximide suppressed the late phase of the PTZ-provoked seizure score significantly only in the null genotype (non-significant in wild type and heterozygote); (PMID 32000863)
- ◐ WWOX clinical case report (uncontrolled) *need-level observation:* Clinician-reported improvement in seizure control on the ketogenic diet in three named patients of a five-patient series: on the delivered text surface the Discussion sentence and Table 1 carry no count, score or instrument (the per-patient narratives were … (PMID 36537114)
- ◐ WWOX clinical case report (uncontrolled) *need-level observation:* Vigabatrin reports are mixed: in one report spasms resolved in the first of two WWOX-null children and

the second child had a partial response with residual cluster spasms at 11 months; (PMID 30361190)
- ◐ WWOX clinical case report (uncontrolled) *need-level observation:* Cannabidiol (class level) and ACTH/corticosteroid/phenobarbital regimens: clinician-reported control in heterogeneous curated reports; no genotype stratification, no comparator, reading depth queried not read.

Counter-evidence

- *need:* Lithium also suppressed in wild-type mice, so a drug effect in a seizing animal is not a WWOX-specific rescue.
- *need:* No drug-by-genotype interaction test and no concurrent same-day PTZ-alone control in the card's reading.
- *need:* CLAIM 031: cognitive and psychomotor impairment preceded epileptic encephalopathy and did not improve with better seizure control in some cases.
- As written in the card: Pro: in one three-genotype experiment ethosuximide's effect was marked significant in the null panel only. Contra: late-phase restriction; cohort variability of a comparable order to the effect; the wild-type baseline is small but not zero, and lithium was still detected inside the same low range (so a pure floor effect is not established); no interaction test;

Limits

- Association is not modulation, and modulation is not rescue; a precedent from another disease is not a WWOX result.
- Evidence scopes are kept apart and never pooled; shared sources are not independent confirmations.
- Access categories are US evidence found in the sources on the retrieval dates (not approval, not suitability for any person, not safety): prescription.
- Technical comparison is not efficacy, safety or experimental validation.
- Sources and receipt references are retained, not independently verified.
- Shared sources are not independent confirmations; scores are not pooled.
- Text may cite repository-internal record ids (for example DL-, CLAIM, map records); only the identifiers in the Sources table are public.

Missing proof and next experiment

Missing datum: drug-by-genotype interaction test.

Next step: Is the late-phase ethosuximide effect in the null reproduced, with a drug-by-genotype interaction test, a concurrent same-day PTZ-alone control in every cell and a per-genotype baseline stated? The R2 design sketch says power depends mainly on which control cohort is representative, not on N alone.

Sources (public identifiers)

- PMID 30361190 `https://pubmed.ncbi.nlm.nih.gov/30361190/` (observation source in the need record)
- PMID 32000863 `https://pubmed.ncbi.nlm.nih.gov/32000863/` (source of an intervention assertion)
- PMID 36537114 `https://pubmed.ncbi.nlm.nih.gov/36537114/` (observation source in the need record)

- ChEMBL CHEMBL696 https://www.ebi.ac.uk/chembl/compound_report_card/CHEMBL696/ (compound identity)

Generated by repurposing_view.py (schema legend.wwox.repurposing_view/2, shareable build). Shareable build: see the attribution footer of the page for the sources embedded.